NU LIPA

SCHOOL OF ARCHITECTURE, COMPUTING, AND ENGINEERING
CS & IT DEPARTMENT
CTSYSADL – SYSTEMS ADMINISTRATION AND MAINTENANCE
MIDTERM EXAMINATION [HANDS-ON]
3RD TERM | AY 2025–2026

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General Directions

1. Read all instructions carefully before starting the examination.
2. Write your complete name and section on the spaces provided.
3. This is a hands-on laboratory examination. Follow each step accurately and in the correct order.
4. Perform all activities only within the designated virtual environment. Do not attempt any actions on live or external networks.
5. You are NOT allowed to access the internet or any unauthorized resources during the examination.
6. Save your work and take screenshots at each major step as required in the deliverables.
7. Any form of cheating or unauthorized collaboration will result in a grade of zero (0) for this examination.
8. Submit all required outputs before the examination period ends.
9. Raise your hand if you have questions or encounter any technical difficulties.

PART 1: Installing and Configuring Active Directory Domain Services

Objectives

Students will:

- Install Active Directory Domain Services (AD DS)
- Promote a server to a Domain Controller
- Create Users and Organizational Units (OUs)
- Join a client computer to the domain

Lab Requirements

Software:

- Windows Server 2012
- Oracle VM VirtualBox

Virtual Machines:

Machine	OS
Server	Windows Server 2019/2022
Client	Windows 10/11

Step-by-Step Instructions

Step 1: Install Windows Server

1. Start the Virtual Machine.
2. Install Windows Server.
3. Configure the following settings:

Computer Name: DC1

Static IP Address:

IP Address: 192.168.1.10

Gateway: 192.168.1.1

DNS: 192.168.1.10

Step 2: Install the AD DS Role

1. Open Server Manager.
2. Click Add Roles and Features.
3. Select Active Directory Domain Services.
4. Install all required features.
5. Wait for installation to complete, then click Close.

Step 3: Promote to Domain Controller

1. In Server Manager, click the notification flag and select Promote this server to a domain controller.
2. Choose Add a new forest and enter the domain name: school.local
3. Configure DNS, Global Catalog, and set a Directory Services Restore Mode (DSRM) password.
4. Complete the wizard and allow the server to restart automatically.

Step 4: Create Organizational Units (OUs)

1. Open Active Directory Users and Computers from Server Manager > Tools.
2. Right-click the domain (school.local) and select New > Organizational Unit.
3. Create the following OUs:
   - Students
   - Teachers
   - IT Department

Step 5: Create User Accounts

1. Inside each OU, right-click and select New > User.
2. Create at least one user per OU. Examples:
   - Juan Dela Cruz (Students)
   - Maria Santos (Teachers)
   - Pedro Reyes (IT Department)
3. Set a password for each user and configure login options as required.

Step 6: Join the Client Computer to the Domain

1. On the Windows 10/11 client VM, open System Properties.
2. Under Computer Name, click Change.
3. Switch from Workgroup to Domain and enter: school.local
4. Log in using the Domain Administrator credentials when prompted.
5. Restart the client computer to apply changes.

PART 2: Configuring DNS and DHCP in a Domain Controller

Objectives

Students will:

- Install DNS and DHCP roles on the Domain Controller
- Configure DNS forward and reverse lookup zones
- Configure a DHCP scope and activate it
- Verify network services from the client machine

Lab Environment

Machine	OS
Server	Windows Server 2019/2022
Client	Windows 10/11

Platform:

- Oracle VirtualBox / VMware

Step-by-Step Instructions

Step 1: Install DNS and DHCP Roles

1. Open Server Manager and click Add Roles and Features.
2. Select and install the following roles:
   - DNS Server
   - DHCP Server
3. Complete the installation wizard and click Close.

Step 2: Configure DNS

1. Open DNS Manager from Server Manager > Tools.
2. Create a Forward Lookup Zone named lab.local.
3. Add a Host (A) Record:
   - Name: server1
   - IP Address: 192.168.1.10
4. Create a Reverse Lookup Zone for the 192.168.1.x subnet.

Step 3: Configure DHCP

1. Open DHCP Manager from Server Manager > Tools.
2. Right-click the server and select New Scope. Configure the following:
   - IP Range: 192.168.1.100 – 192.168.1.200
   - Subnet Mask: 255.255.255.0
3. Set DHCP Options:
   - Default Gateway (Router)
   - DNS Server: IP of the Domain Controller

4. Activate the scope.

Step 4: Configure and Verify from Client

1. On the Windows client VM, set the network adapter to Obtain an IP address automatically.
2. Connect the client to the same network as the server.
3. Open Command Prompt and verify using the following commands:
   - ipconfig (to confirm the assigned IP address)
   - ping server1.lab.local (to verify DNS resolution)
4. Confirm that:
   - The client receives an IP address from the DHCP scope.
   - DNS successfully resolves domain names.
   - The client can communicate with the Domain Controller.

Required Deliverables

Students must submit screenshots for all the following:

Part 1 – AD DS:

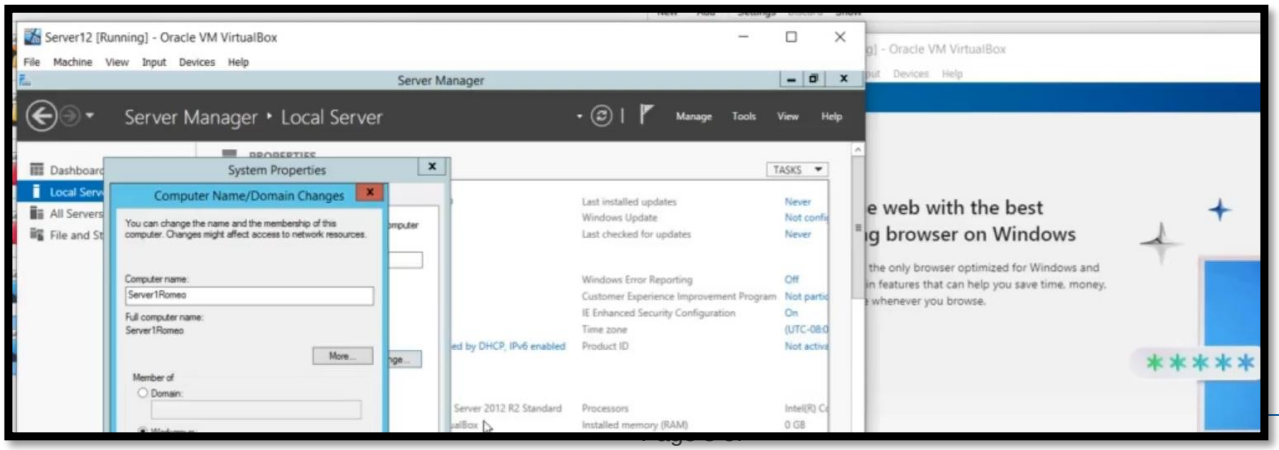
1. Screenshot of AD DS role installation completed.
2. Screenshot of Domain Controller promotion (school.local).
3. Screenshot showing the created Organizational Units.
4. Screenshot of created user accounts assigned to their respective OUs.
5. Screenshot of the client computer successfully joined to school.local.

Part 2 – DNS & DHCP:

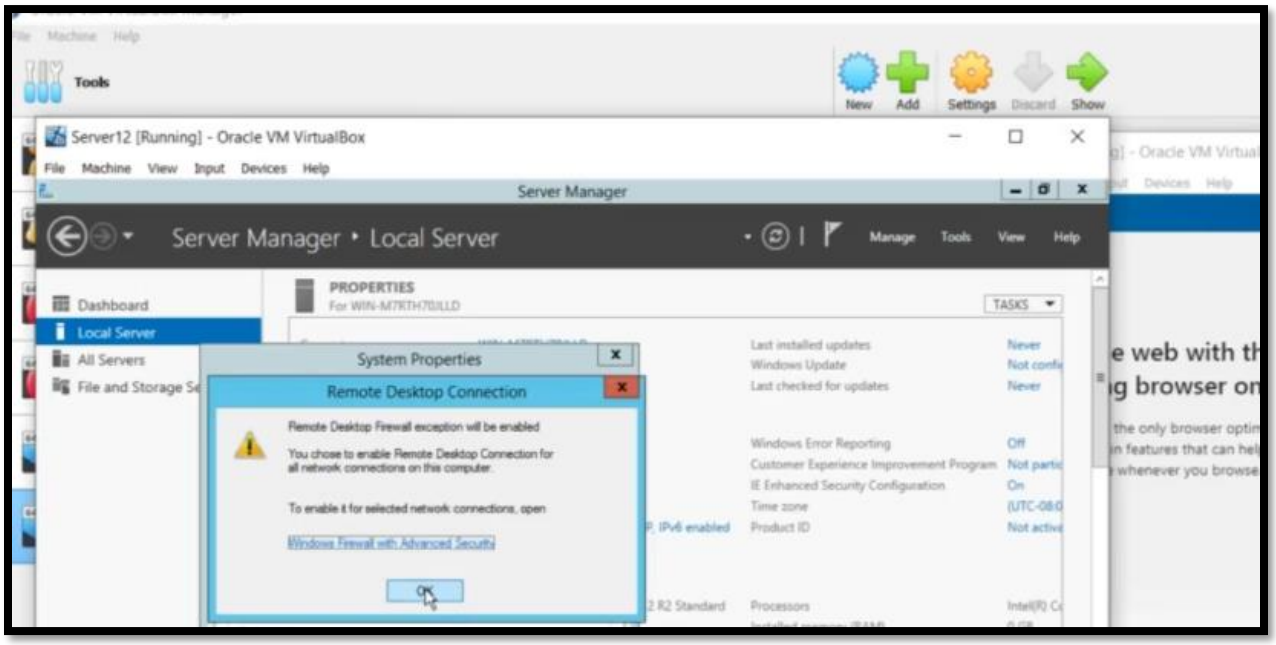
6. Screenshot of DNS Forward Lookup Zone (lab.local) and A Record.
7. Screenshot of DNS Reverse Lookup Zone configured.
8. Screenshot of DHCP Scope activated with correct range and options.
9. Screenshot of ipconfig output on the client showing DHCP-assigned IP.
10. Screenshot of successful ping to server1.lab.local from the client.

DOCUMENTATION:

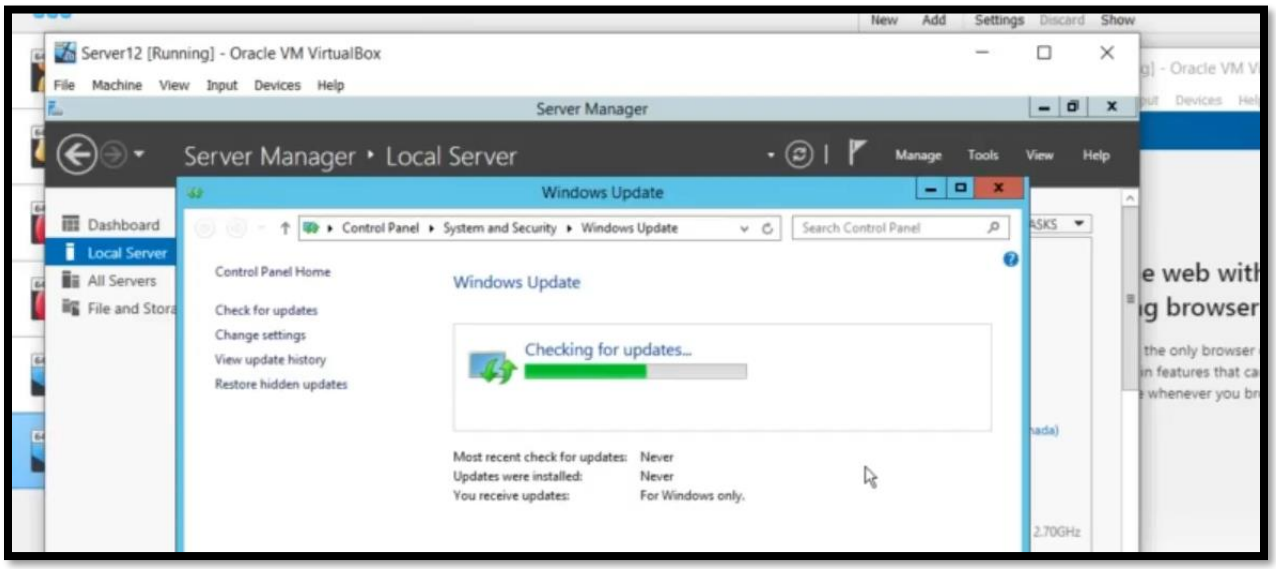
SET UP OF THE SERVER NAME



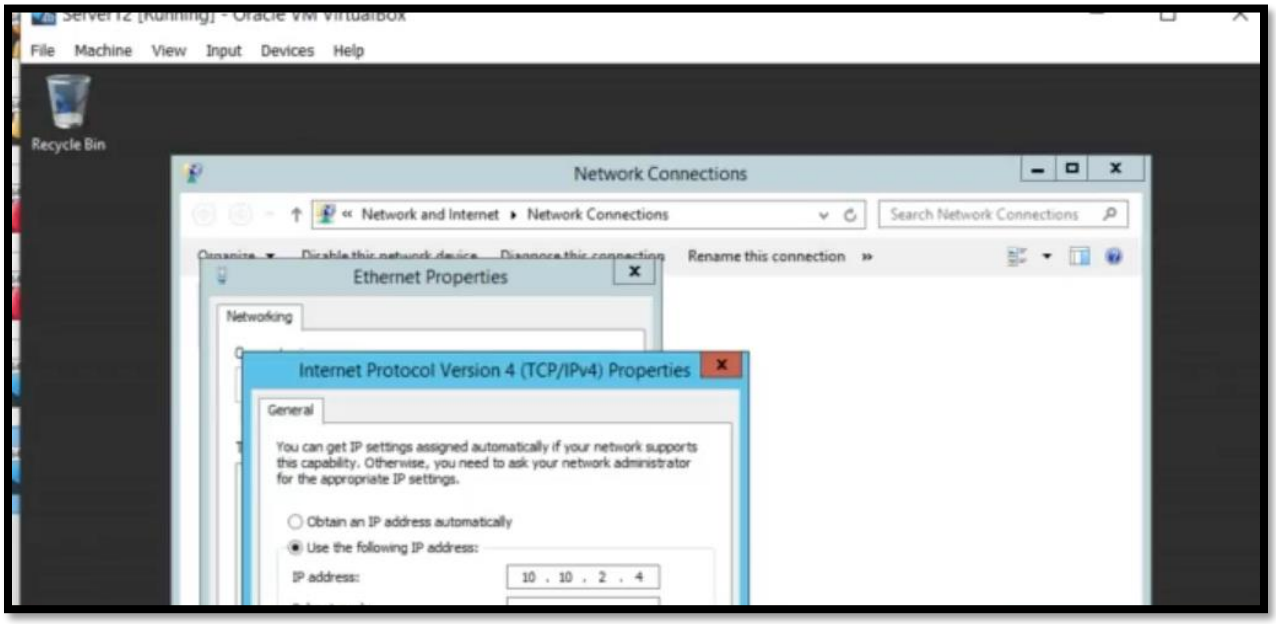
REMOTE DESKSTOP CONNECTION



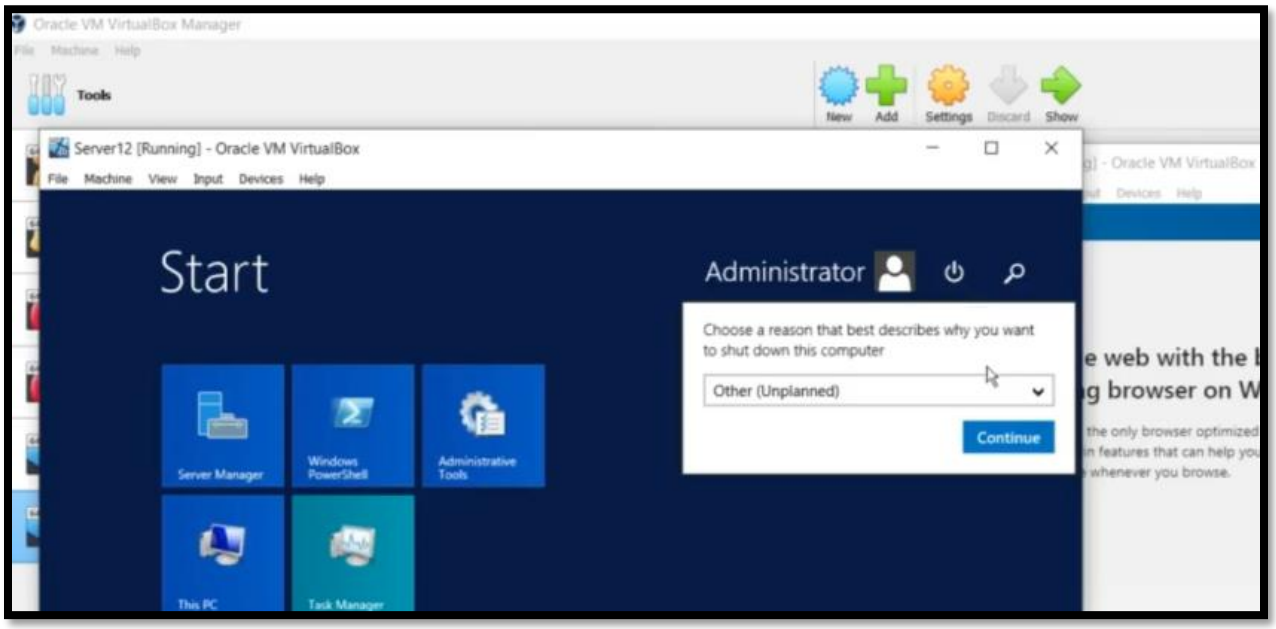
AUTOMATIC UPDATE ALLOWED:



SERVER IP:



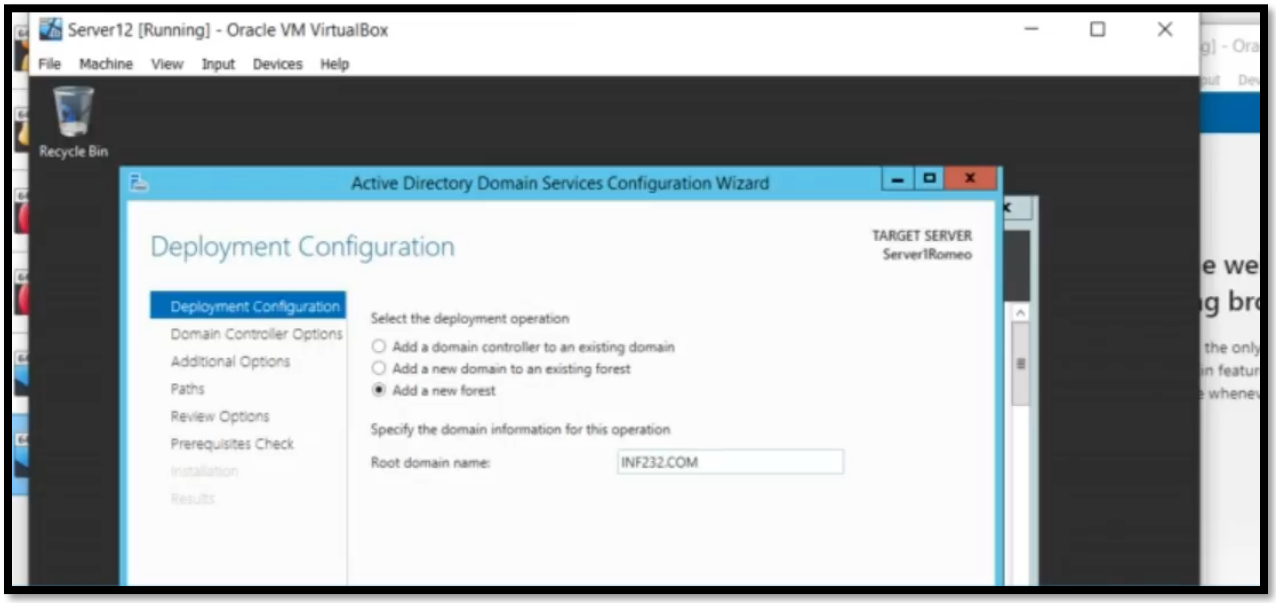
SHUTTING THE ADMIN TO UPDATE THE CHANGES



CHANGING THE ROLES ONING THE DNS SERVER AND DHCP SERVER

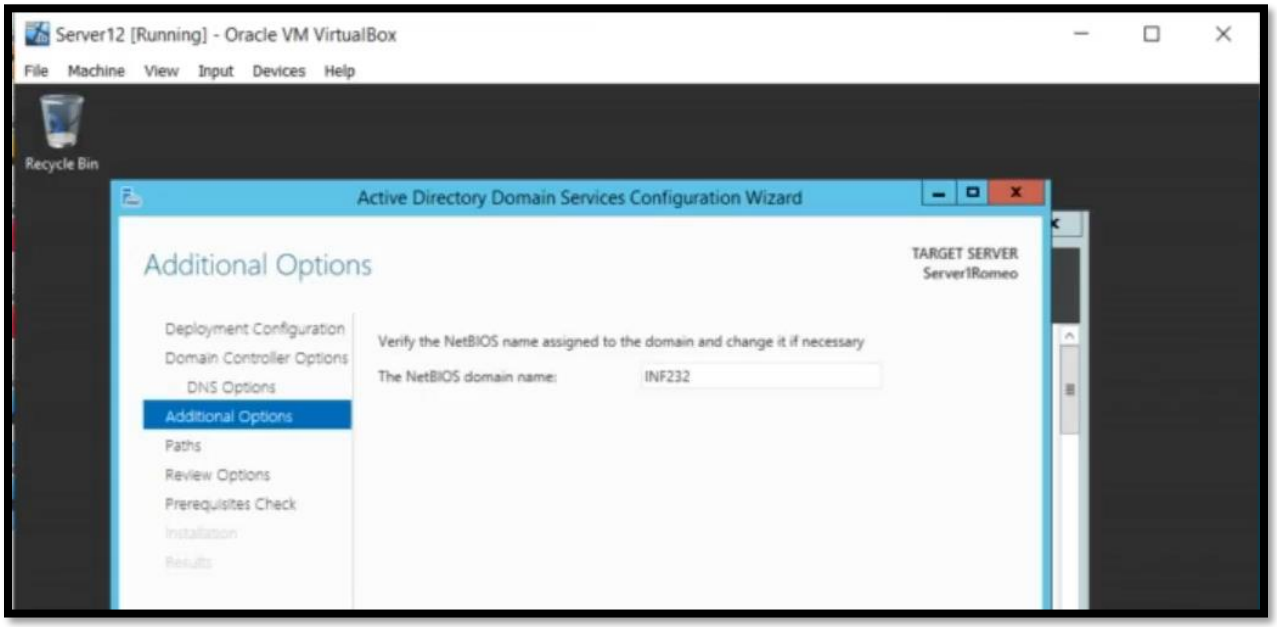
ADD NEW FOREST

PUTTING THE DOMAIN NAME:

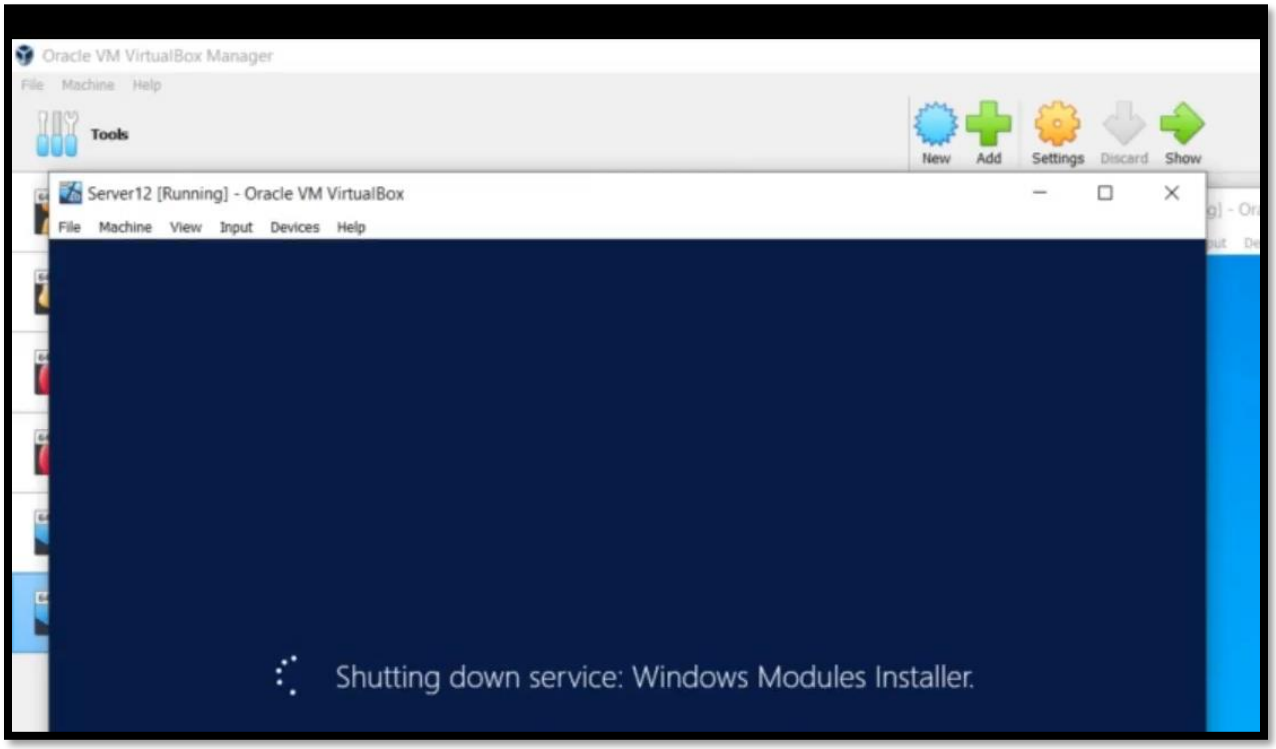


SERVER1ROMEO

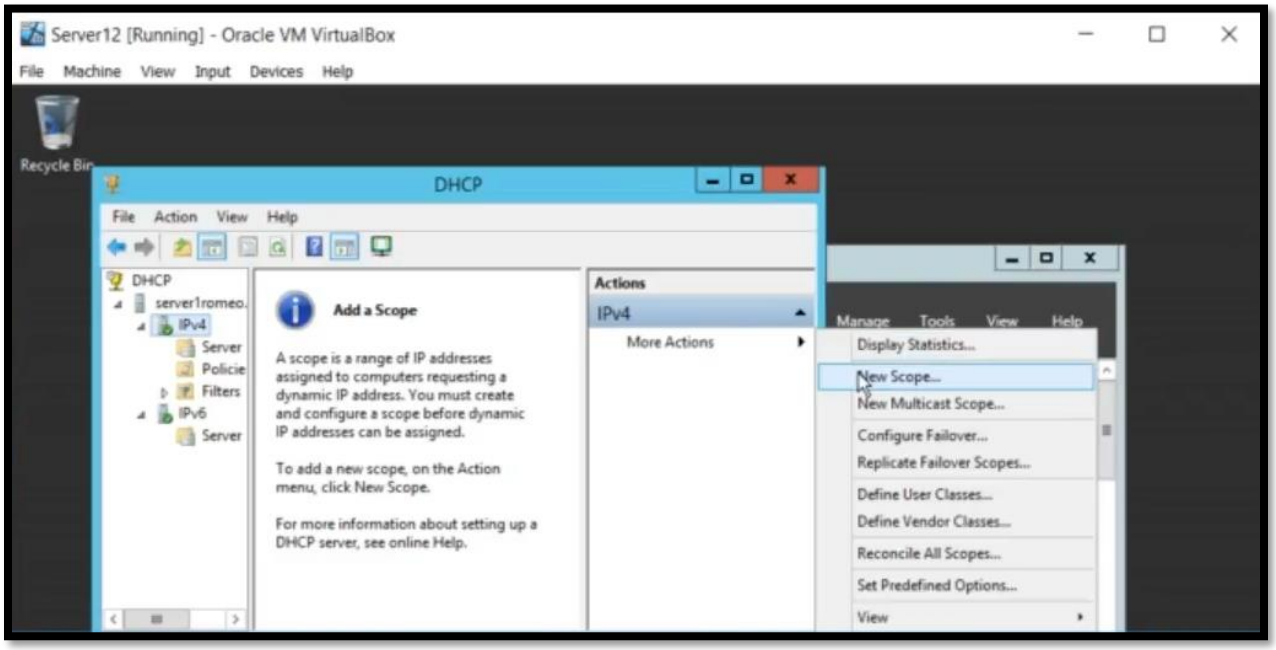
NETBIOS DOMAIN NAME:



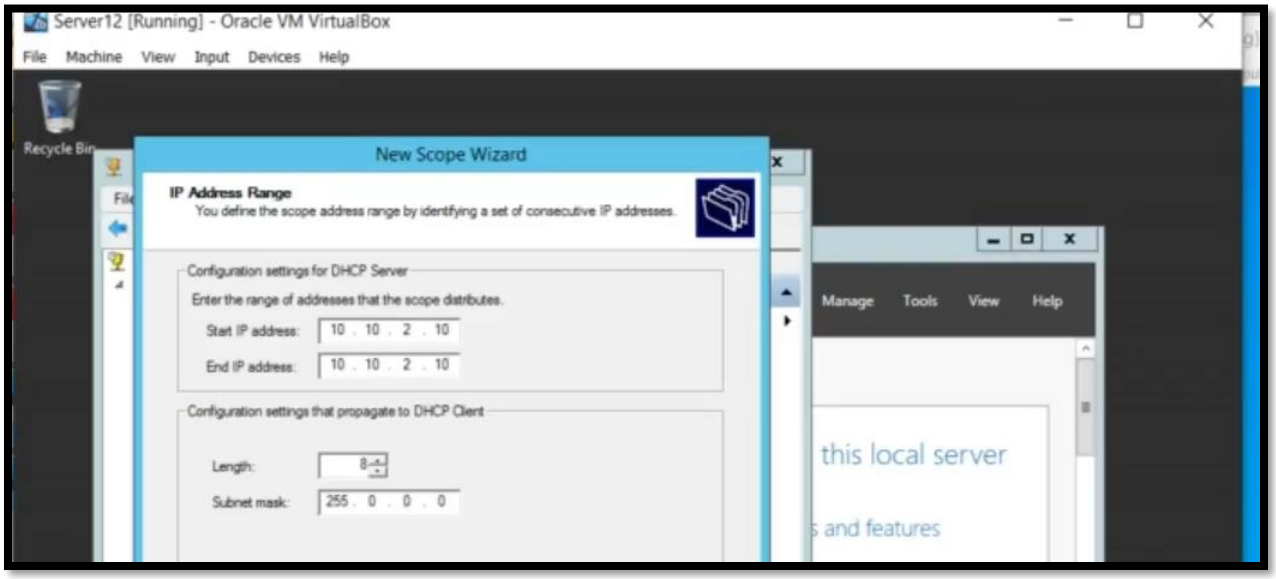
RESTARTING THE SERVER:



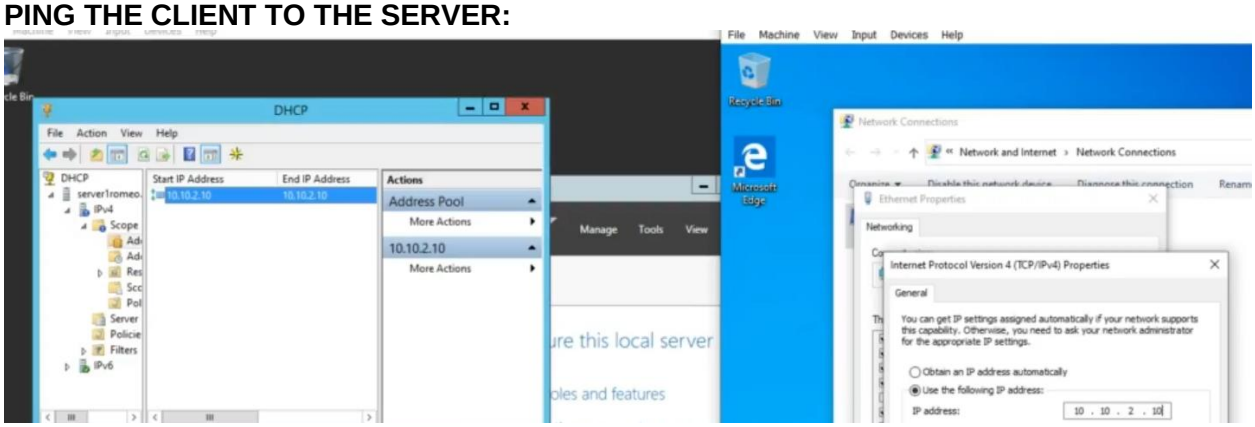
DHCP NEW SCOPE:



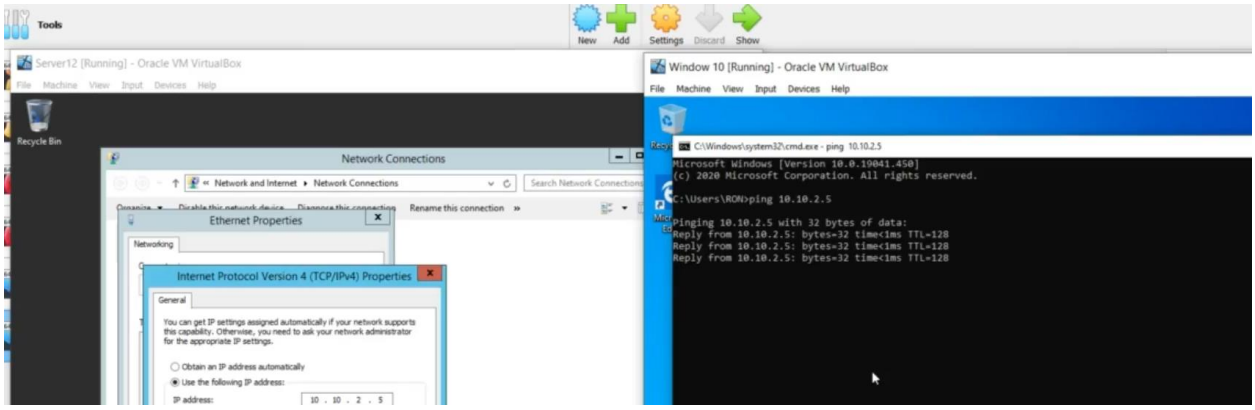
NEW SCOPE WIZARD:

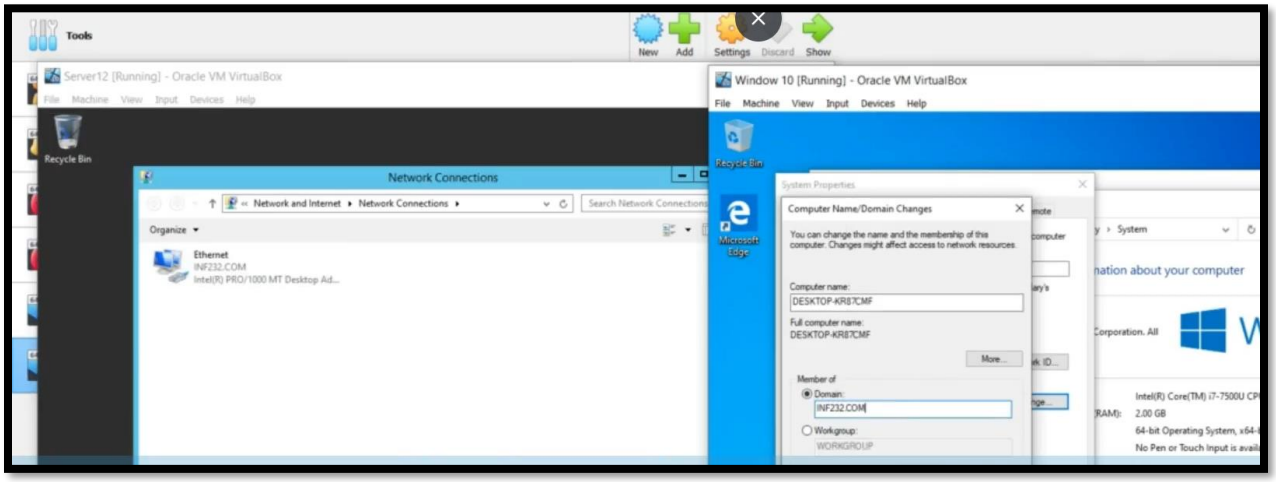


CLIENT SIDE:
PING THE CLIENT TO THE SERVER:

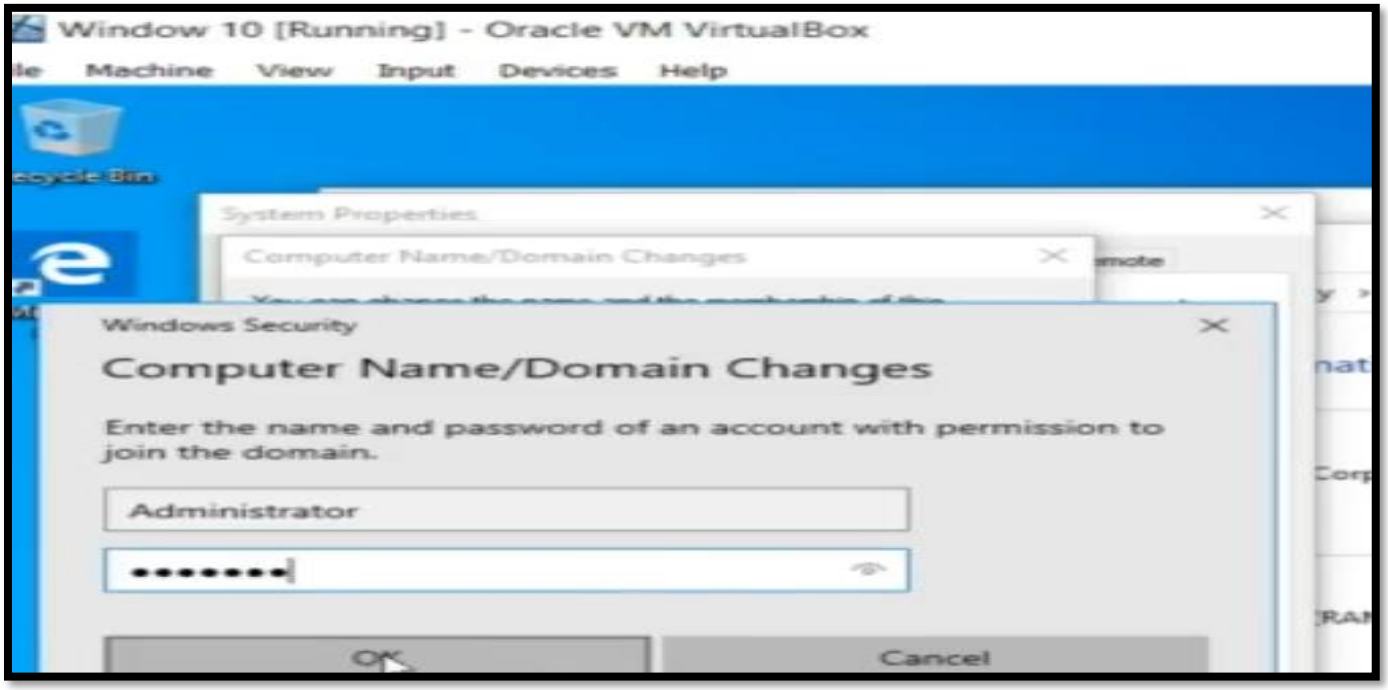


CHECKING IF THE CLIENT CAN FIND THE SERVER:

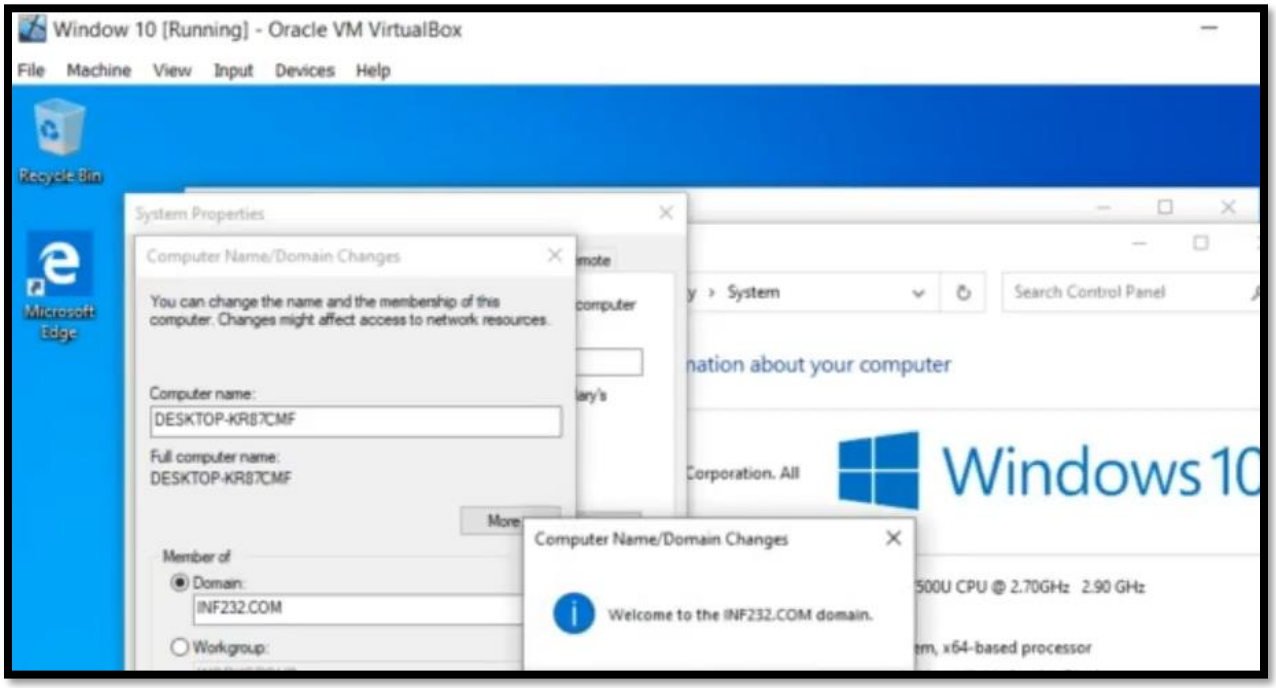




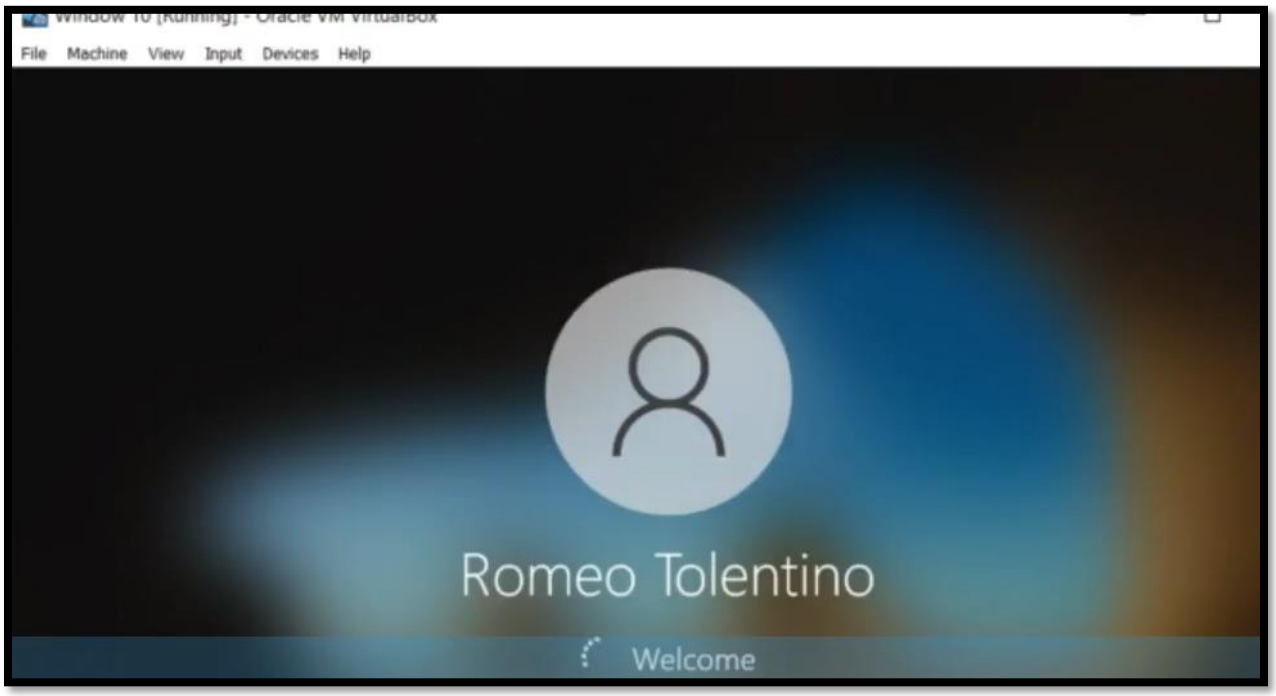
CLIENT COMPUTER NAME AND DOMAIN CHANGES:



WELCOME MESSAGE WHEN YOU CONNECTED THE CLIENT TO THE SERVER



WELCOMING THE CLIENT IN THE DOMAIN SERVER:



Assessment Rubric

Criteria	Description	Points	Score
AD DS Installation	AD DS role was correctly installed on Windows Server with all required features.	15	
Domain Controller Setup	Server was successfully promoted to a Domain Controller with domain name school.local and proper configurations.	15	
Organizational Units	All three OUs (Students, Teachers, IT Department) were created correctly under school.local.	10	
User Account Creation	User accounts were created and correctly assigned to their respective OUs.	10	
Client Domain Join	Windows client was successfully joined to school.local and login via Domain Administrator was demonstrated.	10	
DNS Configuration	Forward and reverse lookup zones were correctly configured with proper A records in DNS Manager.	15	
DHCP Configuration	DHCP scope was correctly configured with valid IP range, subnet mask, gateway, DNS, and the scope was activated.	15	
Client Verification	Client received DHCP IP and DNS resolution was verified using ipconfig and ping commands.	10	
	TOTAL	**100**	